



Environmental Progress Report



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Reflections

Apple’s commitment to climate action has never been more clear.

Thanks to the hard work of teams across our company, we’ve reduced our emissions by over 55 percent since 2015. We’ve crossed key milestones on our environmental journey. And we’re rising to the generational challenge of climate change by working with companies and communities that span the globe.

Apple 2030 is our commitment to be carbon neutral for our entire footprint by the end of the decade. We’ll get there by innovating at every stage of the product lifecycle — from how they’re made, to what they’re made from.

That starts with bringing new clean energy online across our supply chain. Today, more than 320 suppliers have committed to using renewable electricity for Apple production. With over 16 gigawatts already online, they’re avoiding more than 18 million metric tons of greenhouse gas emissions.

Our teams and suppliers are a part of so many communities, so Apple’s work to protect the planet takes us around the world.

That’s why we’re partnering directly with communities and local organizations to support environmental solutions where they’re needed most. In 2023, that meant expanding our Power for Impact program to bring clean energy to underserved communities in Nepal and Colombia, and supporting programs that increase access to safe water and sanitation in India.

We’re also building our products with more recycled and renewable material than ever — work that helps to drive down our carbon footprint. Our customers play an important role in this effort. When you participate in the Apple Trade-In program, you’re helping to bring us closer to the day our products will be made without taking anything from the Earth.

The proof is in our products. Last year, more than 20 percent of the materials we shipped in Apple products came from recycled sources. MacBook Air is our first product made with more than 50 percent recycled material, and we’re making important strides across all of our products. The iPhone 15 lineup, Apple Watch Series 9, and Apple Watch Ultra 2 use 100 percent recycled cobalt in their batteries. More than 99 percent of the tungsten in our products comes from recycled sources. And we’re driving innovation around the world to improve the way our industry recycles and recovers materials.

We’re also finding new and better ways to get Apple products into people’s hands. It’s not just trains, planes, and automobiles — increasingly, we’re shipping Apple products on the open seas, because ocean freight can reduce emissions by as much as 95 percent compared to air transport. And we continue to restore our planet’s ecosystems — like the Atlantic Forest, which stretches from Brazil to Paraguay — that play a key role in removing carbon from the atmosphere.

What does all this work add up to? Progress. And that progress is sorely needed, because the impacts of climate change are all around us. Its frontlines are not in boardrooms or government buildings — they’re in communities. And we owe it to our global community to rise to the challenge of climate change with all the innovation, empathy, and commitment we can muster.

Lisa Jackson
VP, Environment, Policy and Social Initiatives



Report highlights

Reduced our overall emissions by more than 55 percent

We reduced our overall greenhouse gas emissions across scopes 1, 2, and 3 by more than 55 percent compared with our 2015 baseline year — not including offsets.¹ In that same time period, revenue grew by 64 percent. We estimate that we’ve avoided 31 million metric tons of emissions through reduction efforts like transitioning our supply chain to renewable electricity and sourcing recycled content.

Read more on [page 14](#).

Introduced our most significant product emissions reductions to-date

Innovations in design and clean energy drove dramatic reductions in greenhouse gas emissions for the fall Apple Watch lineup — the most significant from our business-as-usual scenario.² This included 100 percent clean energy for manufacturing and product use, 30 percent recycled and renewable material by weight, and 50 percent shipping without the use of air transportation.³

Read more on [page 9](#).



Advocated for reporting transparency

As part of our commitment to greenhouse gas emissions disclosures, we endorsed the landmark California Climate Corporate Data Accountability Act (CA SB 253) to improve transparency and drive progress in the fight against climate change.

Read more on [page 71](#).

Increased our use of recycled materials

We’re making steady progress on our journey toward using only recycled and renewable materials in our products. In 2023, 22 percent of the materials we shipped in Apple products came from recycled sources.⁴ This includes, 99 percent of tungsten, 71 percent of aluminum, 52 percent of cobalt, 25 percent of gold, and 24 percent of lithium in our products.⁵

Read more on [page 17](#).



MacBook Air is the first Apple product made with 50 percent recycled content

This includes 100 percent recycled aluminum in the enclosure, 100 percent recycled rare earth elements in all magnets and, in another first for Apple, 100 percent recycled copper in the main logic board.

Read more on [page 16](#).

More than 320 suppliers committed to renewable electricity

As of March 2024, more than 320 suppliers have committed to sourcing renewable electricity for Apple production, representing 95 percent of our direct supplier spend. We’ve accelerated progress with a mandate in our Supplier Code of Conduct for all direct suppliers to transition to renewable energy in the manufacturing of Apple products.

Read more on [page 27](#).



Launched Grid Forecast

In the contiguous United States, we introduced Grid Forecast, a new tool in the Home app on Apple devices. This tool shows customers when cleaner electricity is available from the grid, empowering them with information to help reduce greenhouse gas emissions.

Read more on [page 30](#).

Goals and progress

Emissions

Goals	Progress	Highlights
Become carbon neutral for our corporate operations.	Achieved	We achieved carbon neutrality in 2020 by expanding energy efficiency initiatives, sourcing 100 percent renewable electricity for Apple facilities, and investing in high-quality carbon credits to offset the remaining hard-to-decarbonize corporate emissions.
Achieve carbon neutrality for our entire carbon footprint by 2030 — reducing related emissions by 75 percent compared with 2015. ⁶	Ongoing	Since 2015, we’ve reduced emissions by over 55 percent across our value chain.
Transition our entire value chain, including manufacturing and product use, to 100 percent clean electricity by 2030. ⁷	Ongoing	As of March 2024, more than 320 suppliers have committed to use 100 renewable energy for their Apple production. These suppliers account for 95 percent of Apple’s direct spend for materials, manufacturing, and assembly of our products worldwide. ⁸

Materials

Goals	Progress	Highlight
Use only recycled and renewable materials in our products and packaging, and enhance material recovery.	Ongoing	In 2023, 22 percent of the material we shipped in our products came from recycled or renewable sources, and we also added titanium to our priority materials list in 2023. ⁹
Transition to 100 percent recycled cobalt, tin, gold, and rare earth elements in select components and applications by 2025. ¹⁰	Ongoing	In 2023, 56 percent of the cobalt shipped in Apple-designed batteries came from recycled sources. ¹¹
Remove plastics from our packaging by 2025. ¹²	Ongoing	Our product packaging shipped in 2023 contains only 3 percent plastic — down from 21 percent in 2015. ¹³

Resources

Goals	Progress	Highlight
Replenish all our corporate freshwater withdrawals in high-stress locations by 2030. ¹⁴	Ongoing	We’ve partnered on freshwater replenishment projects resulting in 31.2 million gallons of volumetric water benefits.
Certify all Apple-owned data centers to the Alliance for Water Stewardship (AWS) Standard by 2025. ¹⁵	Ongoing	We’ve certified five data centers since 2021, and 20 suppliers since 2017 to the AWS Standard.
Expand and grow supplier participation in the Supplier Clean Water Program, prioritizing high water stress locations and driving participants to an average 50 percent water reuse rate by 2030. ¹⁶	New	Through our program, we’ve supported an average 42 percent reuse rate across our 242 participating supplier facilities.
Eliminate waste sent to landfill from our corporate facilities and our suppliers.	Ongoing	Our corporate facilities waste diversion rate increased to 74 percent, driven by progress at our data centers. Throughout 2023, 100 percent of established final assembly sites maintained zero-waste-to-landfill operations.

Overview

Apple 2030

Journey to Apple 2030

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Mapping

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Innovation

Environmental Initiatives



Focus on the future

More than 38 percent of manufacturing electricity for iPhone 15 Pro and iPhone 15 Pro Max is sourced from our supplier clean energy projects.

Environmental Initiatives

Apple 2030

Apple 2030 is our commitment to be carbon neutral for our entire carbon footprint. Our journey to 2030 is focused on first reducing our scope 1, 2, and 3 greenhouse gas emissions by 75 percent compared with 2015, and investing in high-quality carbon removal solutions for the remaining emissions.

[Design and materials](#)

[Electricity¹⁷](#)

[Direct emissions](#)

[Carbon removal](#)



Resources

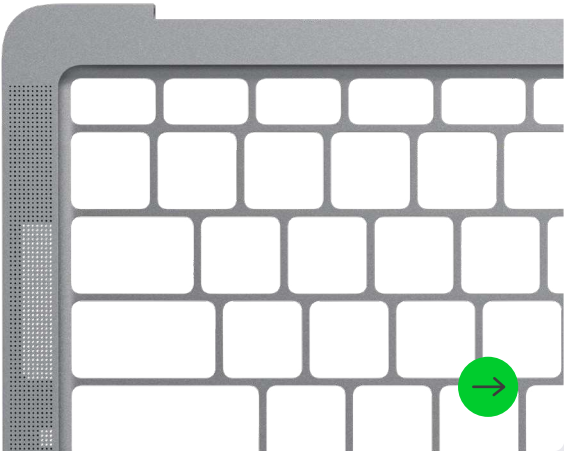
We aim to make durable, long-lasting products and enhance material recovery. And we're committed to stewarding water resources and eliminating waste sent to landfills.

[Product longevity](#)

[Material recovery](#)

[Water](#)

[Zero waste](#)



Smarter Chemistry

Through chemistry innovation and material selection, we design our products to be safer for anyone who assembles, uses, or recycles them — and to be better for the environment.

[Mapping](#)

[Assessment](#)

[Innovation](#)



Apple 2030

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Smarter Chemistry

The time for change
The significant emissions reductions associated with the Apple Watch lineup mark a milestone toward our 2030 goal.

Apple 2030



An ambitious goal for 2030: We committed to be carbon neutral for our entire carbon footprint by the end of the decade.

Our journey to 2030 centers on reducing our scope 1, 2, and 3 emissions first — upstream and downstream — by 75 percent before balancing remaining emissions with high-quality carbon removals.

We've already reduced emissions by more than 55 percent since 2015, even as revenue grew by 64 percent during that same eight-year period.

We're focused on decarbonizing the three largest sources of emissions — materials, electricity, and transportation — across our value chain. We're prioritizing efforts to drastically reduce these emissions with initiatives focused on achieving the broadest impact before applying offsets. For

emissions that can't be further reduced, we primarily look for high-quality, nature-based credits. We invest in carbon sequestration and removal projects to help us achieve carbon neutrality.

Decarbonizing our supply chain is essential to achieving Apple 2030. We've focused on innovative approaches to significantly expand renewable energy across our supply chain and manufacture our products with recycled and renewable materials.



[Apple 2030](#)

[Resources](#)

[Smarter Chemistry](#)

Decarbonizing our value chain

Electricity

Electricity for manufacturing and charging devices represents the largest source of Apple's emissions across all product lines. Achieving carbon neutrality for our entire carbon footprint requires a transition to 100 percent clean energy and significant energy reductions throughout our supplier facilities.

We launched the Supplier Clean Energy Program in 2015 to advance renewable energy throughout our manufacturing supply chain. As of March 2024, more than 320 global suppliers — representing 95 percent of Apple's direct manufacturing spend — have joined the program. These suppliers have committed to using 100 percent renewable electricity for all their Apple production by the end of this decade. The success of this initiative has led to the next phase of our efforts toward achieving a carbon neutral supply chain by 2030.

To rapidly scale and accelerate the progress of the Supplier Clean Energy Program, Apple updated the Supplier Code of Conduct to require all direct suppliers to transition to renewable energy in the manufacturing of Apple products. The codification of this requirement is the next step in our decarbonization journey and a sign of our commitment to integrate supply chain decarbonization into our core business decision-making. We hope that our approach can serve as a model that other companies can follow.

We're also working to address emissions from product use through investments in renewable energy so that every watt of electricity associated with our customers' product use is matched by clean electricity. We're prioritizing product efficiency to reduce the electricity use of our products.

We've also introduced Grid Forecast, a tool in the Home app that informs U.S.-based users when their power grid has cleaner energy available for use. And with iOS 16, we launched the Clean Energy Charging feature in the United States, which looks at the sources of the electricity during expected charge times and optimizes for when the grid is using cleaner energy sources like solar or wind.

Materials

We've advanced toward our 2030 goal by using recycled and renewable materials, which often have a lower carbon footprint than primary materials. And we announced that, by 2025, we plan to use 100 percent recycled cobalt in all Apple-designed batteries, 100 percent recycled tin soldering, 100 percent recycled gold plating in all Apple-designed rigid and flexible printed circuit boards, and 100 percent recycled rare earth elements in all magnets across new products.¹⁸

We've pioneered the use of many recycled materials in our products through world-class product engineering, extensive design qualifications, and supply chain engagement. Our focus is on materials that will yield significant environmental and social

impacts, even when those materials come with notable challenges to building circular supply chains. To read more about our efforts to create circular supply chains, see the [Prioritizing our efforts](#) section. We intentionally design products to minimize the use of primary (non-recycled) materials and maximize recycled content as feasible. And year over year, we innovate and improve manufacturing processes to use materials more efficiently and decrease manufacturing scrap.

In 2023, manufacturing our products accounted for 59 percent of our gross carbon footprint. This includes emissions from fuel combustion; heating, ventilation, and air conditioning (HVAC); refrigeration; use of fluorinated gases; and other physical or chemical processes (excluding transportation). These emissions sources vary widely and require use of diverse technologies and solutions to abate.

Our approach is to identify process emissions across our supply chain, like the use of fluorinated gases in display and semiconductor manufacturing, and launch targeted programs to address these emissions in partnership with our suppliers, governments, and industry stakeholders. We continue to launch supplier programs targeting emissions from manufacturing operations and facilities used to produce Apple products, like the [Supplier Energy Efficiency Program](#) launched in 2015 to help our suppliers optimize their energy use.

Transportation

In 2023, transporting our products to customers accounted for 9 percent of our gross carbon footprint. To address the emissions from transportation, we're shifting more product volume to shipping modes that are less carbon intensive than air transport, such as ocean or rail. Shipping the same Apple product by ocean generates 95 percent fewer emissions than shipping it by air, based on our carbon footprint methodology. We're investigating a transition to low-carbon sustainable aviation fuels (SAF) to reduce the carbon footprint of air shipment. We're also addressing transportation-related emissions through product and packaging design that drives reduced packaging mass and volume, creating boxes that use space more efficiently.

Carbon removals

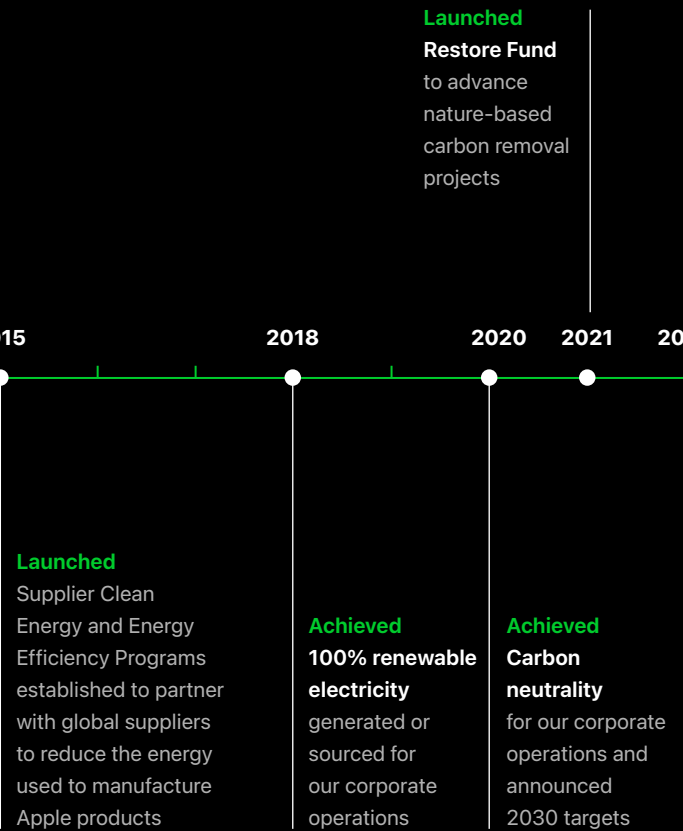
Prioritizing emissions reductions across our entire value chain is critical to limiting climate change to 1.5°C. Still, not all emissions can be avoided or reduced with existing solutions. And some existing solutions will require greater industry and government efforts to scale before broader commercial adoption becomes feasible. As we focus on eliminating the majority of emissions across the value chain, we're also addressing emissions we can't reduce by looking for high-quality carbon credits from nature-based projects. The nature-based projects we select focus

on carbon sequestration, like planting forests and restoring mangroves, and have additional benefits that improve climate adaptation and resilience.

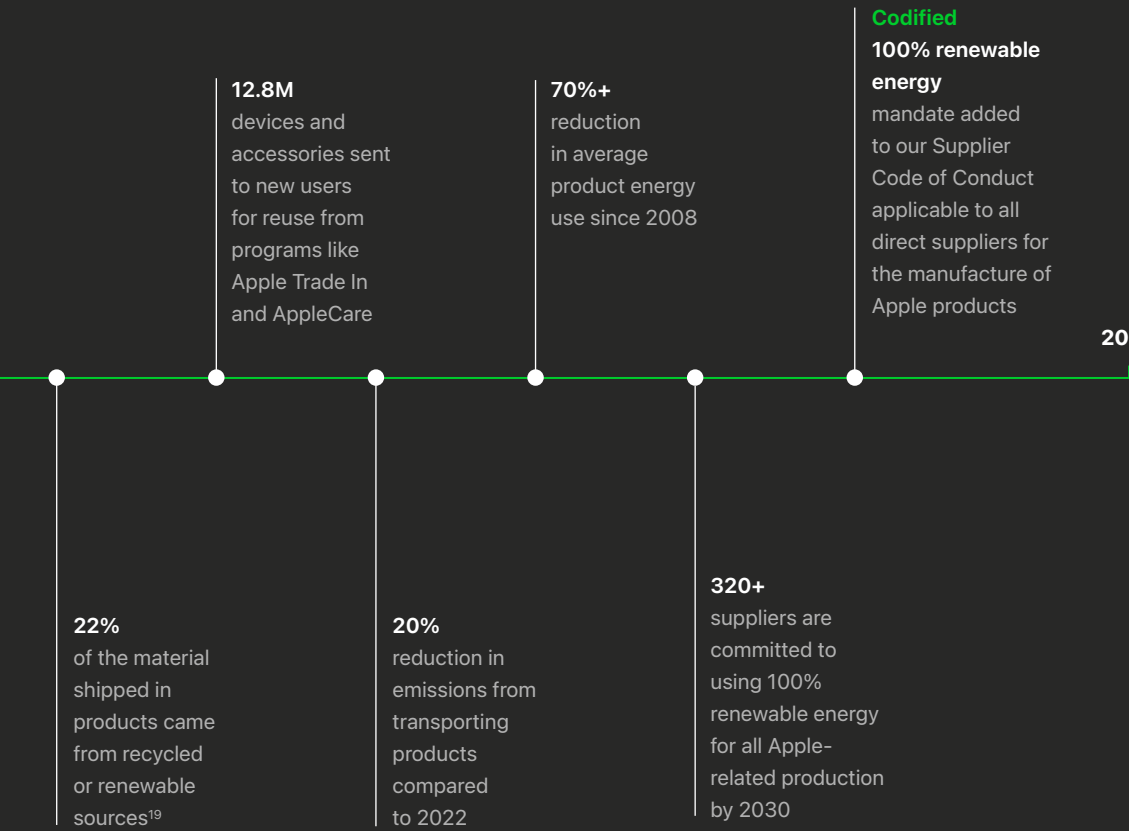
Transparency and integrity are critical to the carbon projects that we fund. The credits from these investments must be additional, permanent, measurable, and quantified — with systems in place to avoid double-counting — and must avoid leakage. To read more about our work in carbon removals, see [Carbon removal](#) and read our white paper, [Apple's Carbon Removal Strategy](#).

Journey to Apple 2030

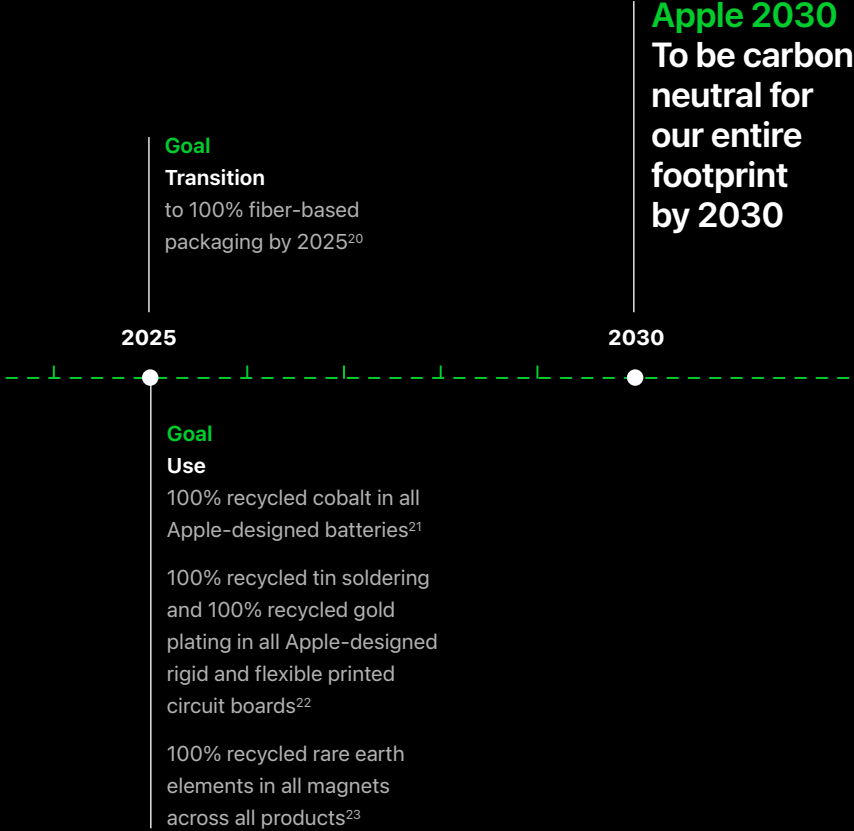
Where we've been



2023 activities



Where we're going



Approach

Apple 2030

We have an ambitious commitment and a science-based plan to reach our Apple 2030 goal. We’re focused on achieving reductions wherever possible, using approaches that offer clear evidence for a way forward while seeking to catalyze industry-wide change.

This begins with working to achieve carbon neutrality across our entire carbon footprint by 2030, setting ambitious targets to reduce our emissions by 75 percent. We prioritize carbon reductions, but for emissions that can’t be mitigated using existing solutions we invest in high-quality carbon removal projects.

Our goal to be carbon neutral extends to our entire carbon footprint and is consistent with the Intergovernmental Panel on Climate Change’s (IPCC) recommendation for global carbon neutrality.²⁴

We’re also committed to working toward reaching a 90 percent reduction in emissions from our 2015 baseline by 2050. Attaining deep decarbonization will require a collective, worldwide effort. Entire industries and economies must decarbonize. And while reaching a 90 percent reduction in emissions is outside Apple’s or any one company’s control, we’re committed to taking actions that support this goal.

We’ve made significant progress by cutting emissions across our value chain by more than 55 percent since 2015. This started with our transition to sourcing 100 percent renewable electricity at our offices, retail stores, and data centers, which we achieved in 2018. And in 2020, we achieved carbon neutrality for our corporate emissions.²⁵

APPLE 2030 ROADMAP

Addressing Apple’s carbon footprint through four pillars



Design and materials

Designing products and manufacturing processes to be less carbon intensive through thoughtful material selection, increased material efficiency, greater product energy efficiency, the use of recycled and renewable materials in our products and packaging, and enhanced material recovery

>75 percent emissions reduction



Electricity

Increasing energy efficiency at our facilities and in our supply chain, and transitioning the electricity in our entire product value chain — including manufacturing and our customers’ product use — to 100 percent clean electricity by 2030



Direct emissions

Reducing direct greenhouse gas emissions in our facilities and our supply chain through process innovation, emissions abatement, and shifting away from fossil fuels



Carbon removal

In parallel with our emissions reduction efforts, scaling up investments in carbon removal projects, including nature-based solutions that protect and restore ecosystems around the world

<25 percent of footprint

[Apple 2030](#)

[Resources](#)

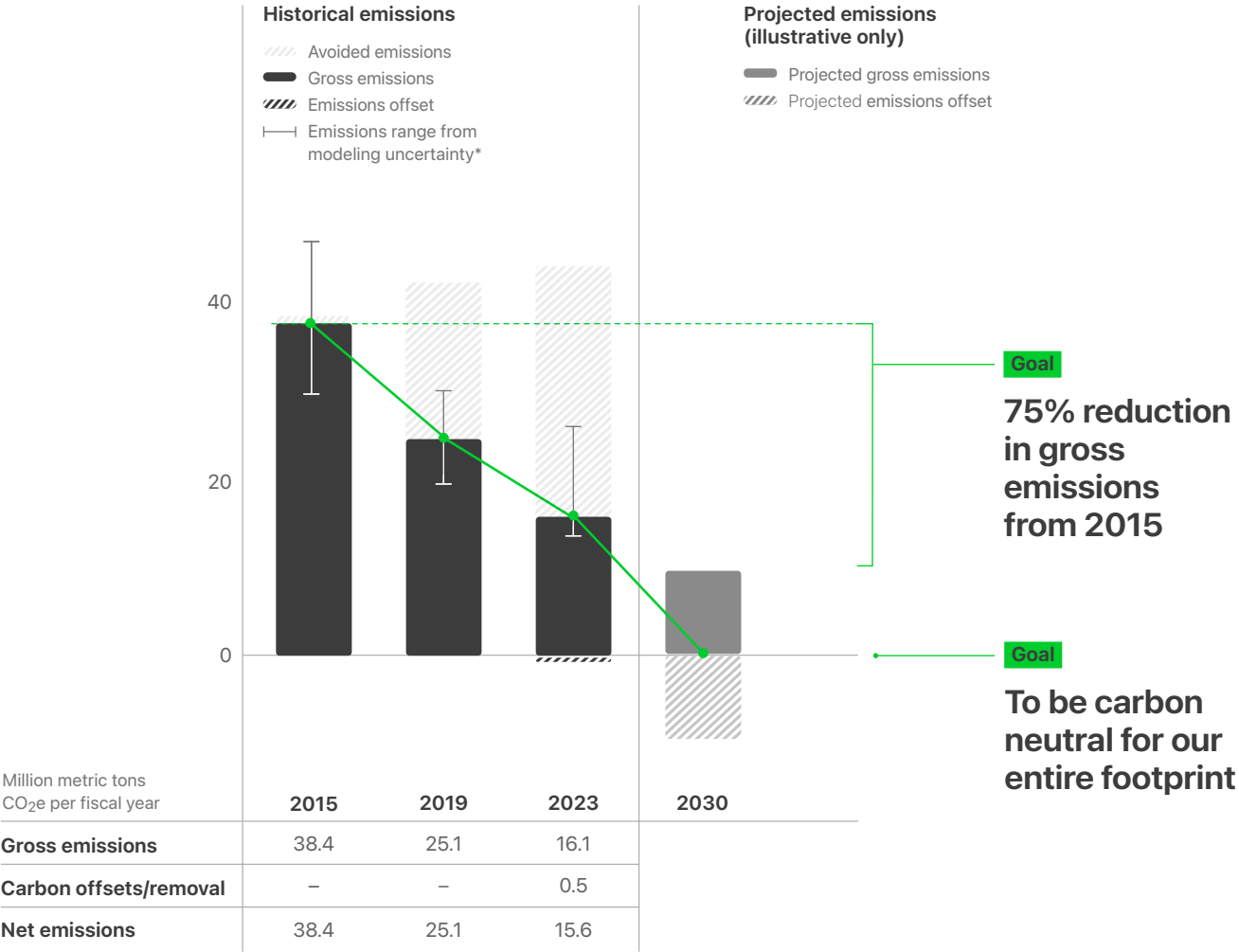
[Smarter Chemistry](#)

Environmental progress can and should be good for business. We underpin our climate strategy with strong business principles and innovation while harnessing the power of markets to replicate our solutions at scale — aiming to create the impact necessary to meet global reduction targets. We’re also working to accelerate the global transition toward decarbonization while exploring the next generation of solutions and advocating for stronger policies. These efforts include fostering new and strengthened collaborations in public and private partnerships and investing in technological advancements. We’ve issued \$4.7 billion in green bonds to model how businesses can drive investments to reduce global emissions.

Clear principles guide this work. Our emissions reduction targets are aligned with what current climate science shows is necessary to help limit warming to 1.5° C. We calculate our footprint across our value chain to include both direct and product-related emissions — from sourcing materials through end of life. We use the results of our detailed carbon accounting to adjust our Apple 2030 roadmap, which lays out our plan to become carbon neutral. And we’re committed to disclosing our carbon footprint as well as our climate strategy and progress. This means sharing both challenges and successes. Our annual Environmental Progress Report, as well as our response to the global disclosure nonprofit CDP, provide details on our progress.

Transparency and disclosure are essential not only for sharing our climate strategy and progress but also for sending clear signals and inviting others to work with us. And progress needs to include low-income and historically marginalized communities that too often bear the brunt of climate change. We’re pursuing ways to directly support these communities in our climate programs. To learn more about our work, read our [feature](#) on our Power for Impact program and our [feature](#) on our Impact Accelerator program.

APPLE’S PROGRESS TOWARD CARBON NEUTRALITY



To see our progress toward our 2030 goal, see our [Journey to Apple 2030 timeline](#).

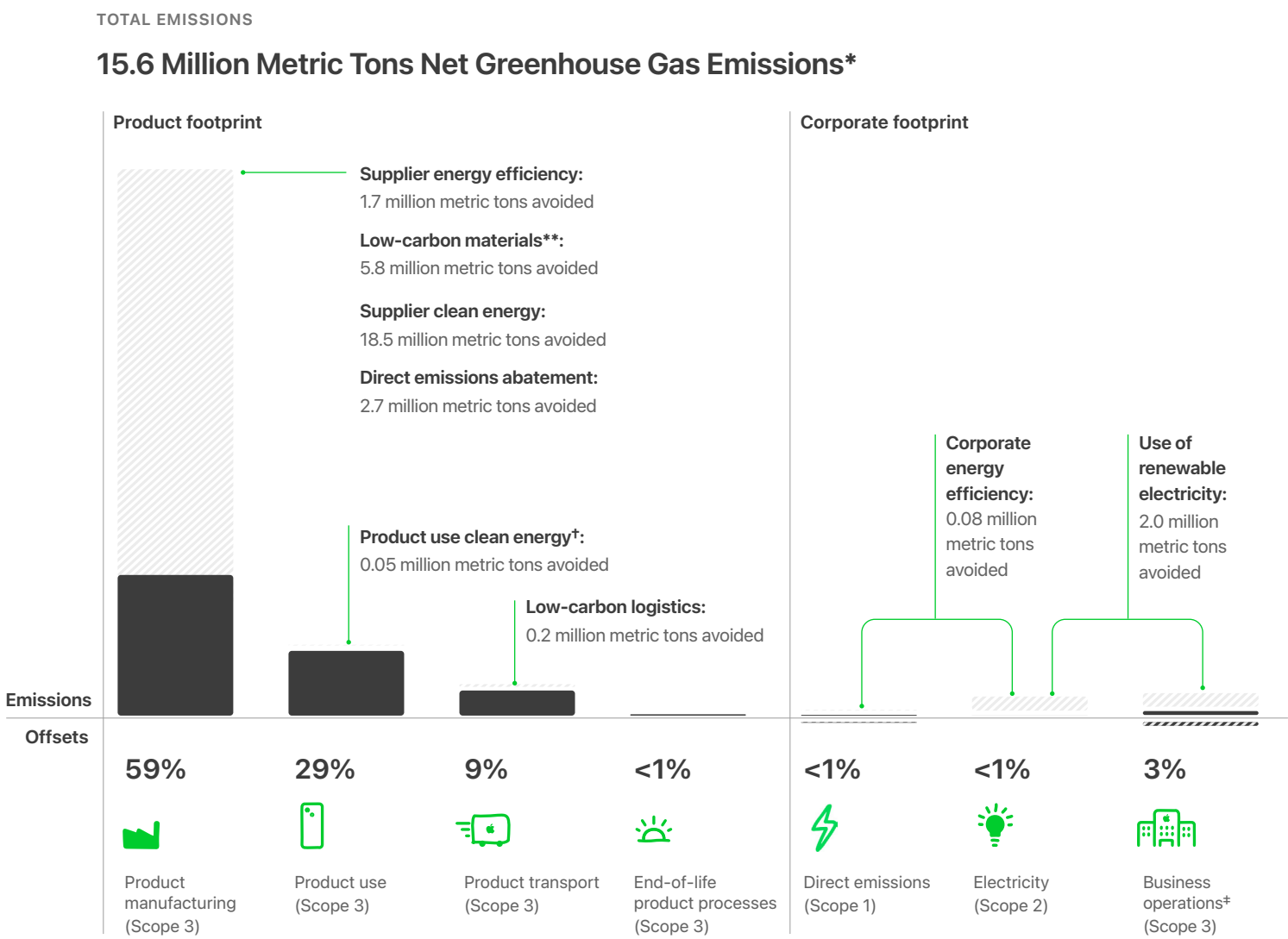
In our latest [Annual Green Bond Impact Report \(PDF\)](#), we share progress on the projects funded in 2023.

Read our latest response to the [CDP Climate Change 2023 questionnaire \(PDF\)](#).

* Error bars: We’re continuously refining our methodology to improve our carbon footprint estimate. But there’s inherent uncertainty in modeling product-related greenhouse gas emissions, as illustrated by the error bars in this graphic.

Apple's comprehensive carbon footprint

In 2023, we estimate that our environmental programs avoided 31 million metric tons of emissions across all scopes. Initiatives that we've been growing for years continue to yield clear results, including sourcing 100 percent renewable energy for our facilities, transitioning suppliers to renewable energy, and using low-carbon materials in products.²⁶ While our revenue has grown by more than 64 percent since 2015, our gross emissions have decreased by more than 55 percent.



////// Avoided emissions

■ Gross emissions

////// Emissions offset

* Net greenhouse gas emissions represents our total gross footprint minus carbon offsets. Percentages shown for each emissions category represent the share of Apple's gross footprint. Totals add up to more than 100 percent due to rounding.

** Low-carbon materials represents emissions savings from transitioning to recycled materials in our products and using low-carbon aluminum, as described on page 33.

† Clean energy represents savings from clean energy procured by Apple or its suppliers.

* Business operations includes business travel, commute, work from home, upstream impacts from scope 1 fuels, and third-party cloud services.

Developing new approaches to disassembly and recovery

Innovation plays a key role in realizing the potential of recycling Apple products. We continue to develop better, more efficient means of disassembling products that maximize material recovery while minimizing waste. Our Material Recovery Lab (MRL), an R2-certified facility in Austin, Texas, focuses on assessing the recyclability of our products, helping inform design decisions that support disassembly and recovery. The MRL's work has led the way in automated approaches to material recovery with our robots Daisy, Dave, and Taz.

We're also developing new technologies at our R2-certified asset recovery center in Santa Clara Valley, California, that uses automation and machine learning to facilitate recycling processes that typically require manual sorting, which can be time intensive and error prone. In 2023, we deployed autonomous mobile robots (AMRs) to transport materials across our operations, as well as an automated product sorter that uses machine learning to identify and sort our products into 17 categories. These technologies improve material handling and sorting efficiency, which has a direct impact on the material capacity of our operations. This work aims to create low-cost solutions that our suppliers can deploy to recover more materials, freeing up their employees for more complex tasks that can't be easily automated.

We're also working to improve the material handling efforts of demanufacturing. Last year, we began deploying overhead projector-based augmented reality systems to recyclers, which project instructions for demanufacturing processes onto the work surface. These systems are designed to further improve safety, productivity, and efficiency.

Designing for the next generation of recycling

We're committed to a long-term approach to recycling innovation, relying on and continually improving current approaches while nurturing new and emerging technologies. And we continue to support initiatives that set out to redefine disassembly and material recovery.

We're investing in recycling innovations that can have industry-wide impacts. For several years, we've worked with Carnegie Mellon University's Biorobotics Lab in the School of Computer Science's Robotics Institute to identify and disassemble e-scrap. This project has the potential to enable recyclers to recover the materials at a higher quality. The software we develop will be open source and available to others in the industry who are working to maximize the recovery of recyclable materials. We've also collaborated with the lab to improve the accuracy of disassembly and the recovery of

materials through X-ray imaging combined with RGB imaging. And we're optimizing our robots' abilities to handle objects and adjust to contact, enabling them to "feel" by using high-frequency force feedback and machine learning. Through this work, we're driving toward developing truly intelligent disassembly technology. Learn more about Apple's research with Carnegie Mellon University in our [white paper](#).

Advanced disassembly

Daisy: In 2023, we expanded the capabilities of Daisy, which can now disassemble 29 iPhone models into discrete components. Daisy can disassemble up to 1.2 million phones each year, helping us recover more valuable materials for recycling. From just one metric ton of iPhone main logic boards, flexes, and camera modules recovered by Daisy, our recycling partner can recover the same amount of gold and copper as can be obtained from more than 2,000 metric tons of mined rock.⁶⁸ The U.S. Patent and Trademark Office issued Apple five patents related to Daisy, and we're willing to license these to researchers and other electronics manufacturers who are developing their own disassembly processes in hopes that they will follow suit and work to increase the share of recycled materials in the supply chain.⁶⁹



As of 2024, Taz is operating at our recycling partner in China.

Dave: A robot that enables the recovery of rare earth magnets as well as tungsten and steel by specializing in the disassembly of the Taptic Engine. As of 2023 Dave is operating at our recycling partner in China.

Taz: Helps recycle modules by separating magnets containing rare earth elements from audio modules. While rare earth elements are typically lost in conventional shredders, Taz is designed to access these valuable materials, improving our overall recovery rate. As of 2024, Taz and Daisy are operating at our recycling partner in China.

Water

We aim to advance water security and protect communities in the places where we and our suppliers operate through actions that improve freshwater availability, quality, and equity.

Strategy

Water is a local resource, so our strategy is context based and respectful of local conditions where we and our suppliers operate. We collect and analyze data and site-level feedback to make sure we understand our water impacts. We use tools like the World Resources Institute (WRI) Aqueduct Water Risk Atlas to gain insights on local watershed health, such as baseline water stress.

We’ve developed a water footprint to understand how we use water across our value chain — in our own operations, our services, and our manufacturing supply chain. Through our water footprint efforts, we’ve identified that 70 percent of our corporate water use occurs in areas with high or extreme basin stress, and that the majority of our water-related impact is in the manufacturing supply chain.⁷⁰

We use these insights to inform our water program priorities. And we work directly with our suppliers to ensure that they have robust policies, are managing their wastewater systems efficiently, and are reducing their overall consumption by reusing wastewater. We partner with over 240 supplier facilities to guide them toward world-class water conservation and management.

We know that we can’t address water stewardship challenges alone. Making a meaningful impact on water requires collective action. To do that, we must go beyond our own operations to collaborate with communities and work in basins on stewardship, replenishment, and WASH (water access, sanitation, and health) projects throughout our value chain.

- We focus our work on five strategic pillars:



Low-water design

Minimizing water impacts in the design of products, services, and sites



Site efficiency and conservation

Improving the performance of existing sites and processes



Site-level water stewardship

Demonstrating responsibility beyond our facilities through watershed-level management



Replenishment and nature-based solutions

Improving water availability, quality, and access through regenerative approaches



Leadership and advocacy

Advancing water management through policy, advocacy, and technology innovation
- ## 2023 progress
- 31.2M gallons
- We’ve collaborated with partners on freshwater replenishment projects resulting in 31.2 million gallons of volumetric water benefits.
- 20 suppliers
- We’ve certified 20 suppliers and five data centers to the AWS Standard.
- 42% reuse rate
- Through our Supplier Clean Water Program, we’ve supported an average 42 percent reuse rate across our 242 participating supplier facilities.

Low-water design

We approach low-water design by first focusing on site selection, conducting a water risk evaluation to determine whether a potential site is in a water-stressed area. The results inform our site selection and help mitigate the impact of our expected water use. We then design solutions to manage the quality of the wastewater that we return to the watershed.

In our corporate operations, we discharge the majority of our wastewater to the sanitary sewer. Up to 54 percent of the water that we use at our corporate locations is discharged back into the local water system rather than consumed.⁷¹ We monitor the water quality from sites where we have an industrial wastewater discharge so that each location meets or exceeds local discharge requirements.

At our Capstone campus in Austin, Texas, we aim to achieve net-zero water use by relying on an onsite wastewater reuse system, as well as condensate recovery and stormwater capture, limiting potable water use to potable purposes only. This effort

will save up to 60 million gallons of freshwater use annually.⁷² Using wastewater for cooling will also allow us to achieve 28 million kilowatt-hour in energy savings annually at the location. In total, our low-water design efforts avoided 150 million gallons of freshwater use in 2023.⁷³

In our supply chain, a majority of water is used during manufacturing. For example, in 2023 we continued an initiative that focuses on eradicating potential hazards associated with liquid discharge throughout our manufacturing supply chain. The purpose of the initiative is to enhance and support our electrochemical metal-finishing processes, like anodizing. This effort incorporates cutting-edge technologies, including optimal water management, modular filtration and recovery units, acid recovery systems, and evaporators. Pilot lines have demonstrated success during the enclosures anodizing process with recovery and reuse rates reaching 75 to 80 percent for water and more than 95 percent for acid. This process improvement has the potential to save hundreds of millions of gallons of water annually when applied at scale.

Site efficiency and conservation

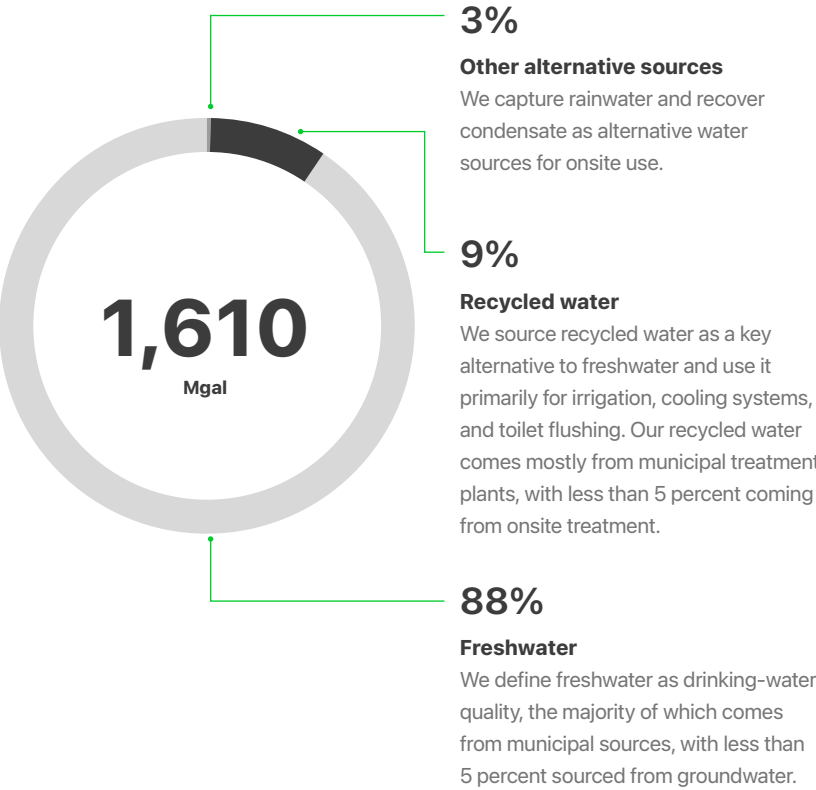
Next, we identify steps to use less freshwater in our existing operations, prioritizing regions where our efforts can immediately reduce stress on local watersheds.

In 2023, our facilities used about 1.6 billion gallons of water in our direct operations, a 5 percent increase from 2022 — driven mostly by increases in our corporate campus portfolio. However, through our site efficiency and conservation efforts, we saved 79 million gallons of freshwater in 2023.⁷⁴ Last year, alternative water sources accounted for about 12 percent of our total corporate water usage — primarily from municipal recycled water sources. Additional efficiency and conservation accomplishments include:

- We used rainwater capture systems to source 9 million gallons of water at Apple locations in Texas, North Carolina, Ireland, Taiwan, and Japan.
- In 2023, we captured 2.6 million gallons of condensate from our condensate recovery systems in Santa Clara Valley and Austin.
- At Apple Park, we completed a new blending tank project that will enable us to replace up to 18 million gallons of potable water with recycled water per year.

WATER USE AT CORPORATE FACILITIES

We track our corporate water use for our data centers, retail stores, distribution centers, and corporate offices



At our data centers, server upgrades in 2023 resulted in 12 million gallons of water saved. We’ve also piloted a resin water treatment system that reduced makeup water use by 30 percent and discharge by up to 60 percent.⁷⁵ Based on this success, we’re implementing this technology at our data centers in Prineville, Oregon, and Mesa, Arizona.

We also continued to pilot a plant-based water treatment using sustainably harvested sphagnum moss at several data centers, helping eliminate the need for biocides and corrosion inhibitors. This system, which uses compostable, natural sphagnum moss to improve water quality, can further support water savings. We implemented this at our Reno, Nevada, and Maiden, North Carolina, data centers and have begun a permanent installation in Mesa, Arizona.

Our supply chain accounts for 99 percent of our total water footprint, based on our detailed water inventory. We require our suppliers to maintain the high standards for water discharge outlined in the Apple Supplier Code of Conduct. Through our Clean Water Program, we help suppliers minimize process water impacts and adopt best practices in water management and wastewater treatment.

Since the program’s launch in 2013, the average reuse rate of the 246 participating suppliers has increased to 42 percent, saving our suppliers 12.7 billion gallons of freshwater in 2023 and a total of over 76 billion gallons of water savings.⁷⁶ These savings come from a range of initiatives, such as the reuse of reclaimed water, upgrades to water-efficient equipment, and countercurrent rinse methods.

Site-level water stewardship

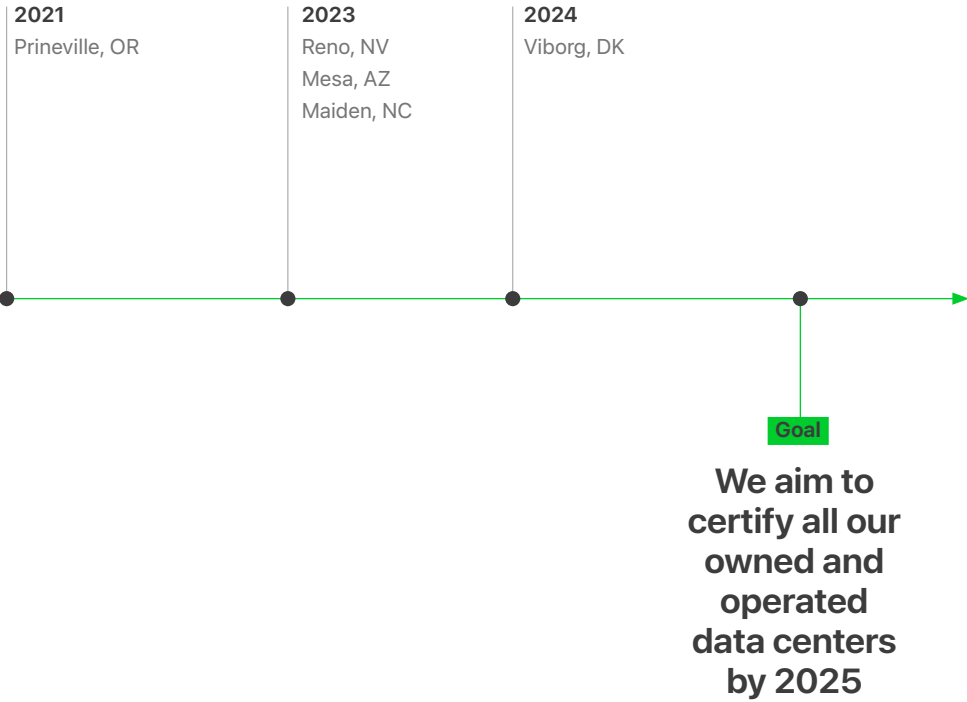
We engage with the communities around our facilities for the protection of local watersheds. Since 2018, we’ve partnered with the Alliance for Water Stewardship (AWS) to advance the AWS Standard, the first global framework to measure responsible water stewardship across social, cultural, environmental, and economic criteria. In 2020, we joined the AWS board of trustees to highlight water stewardship opportunities to our suppliers and promote collective action on shared water challenges impacting the sector.

Through AWS, we’ve certified five of the seven data centers that we own and operate in Prineville, Oregon; Reno, Nevada; Maiden, North Carolina; Mesa, Arizona; and Viborg, Denmark. We’re on track to certify the remaining two Apple-owned-and-operated data centers by 2025.⁷⁷

Since 2018, 20 of our supplier sites have been certified to the AWS Standard, 16 of which achieved a Platinum rating — the highest score within the AWS framework. In May 2023, we partnered with AWS and the Suzhou Industrial Park Administration Committee to conduct a training session for Suzhou Industrial Park (SIP) tenants. This training promoted water stewardship and helped establish a more comprehensive water management system for their factories, including Apple suppliers.

CERTIFIED DATA CENTERS

Number of Apple-owned data centers certified by Alliance for Water Stewardship



Data



Data

Greenhouse gas emissions

We account for our carbon footprint by following internationally recognized standards, like the World Resources Institute (WRI) Greenhouse Gas (GHG) Protocol and ISO 14040/14044.¹ Improving the accuracy of our carbon footprint is an ongoing process — as we learn more, we refine our carbon models and adjust our climate roadmap. We also regularly revisit the boundary of our carbon footprint as our data sources improve and our business evolves.

		Fiscal year				
		2023	2022	2021	2020	2019
Corporate emissions (metric tons CO ₂ e) ²	Gross emissions	324,100	324,000	166,380	334,430	573,730
	Scope 1	55,200	55,200	55,200	47,430	52,730
	Natural gas, diesel, propane	35,300	39,700	40,070	39,340	40,910
	Fleet vehicles	17,000	12,600	12,090	4,270	6,950
	Other emissions ³	2,900	2,900	3,040	3,830	4,870
	Scope 2 (market-based) ⁴	3,400	3,000	2,780	0	0
	Electricity	0	0	0	0	0
	Steam, heating, and cooling ⁵	3,400	3,000	2,780	0	0
	Scope 3	412,800	265,800	108,400	287,000	521,000
	Business travel	225,700	113,500	22,850	153,000	326,000
	Employee commute ⁶	164,100	134,200	85,570	134,000	195,000
	Upstream impacts (scope 1)	18,300	10,600	0	0	0
	Work from home (market-based)	4,700	7,500	0	0	0
	Transmission and distribution loss (market-based)	N/A	N/A	N/A	N/A	N/A
	Third-party cloud (market-based)	0	0	0	0	0
Product life cycle emissions (metric tons CO ₂ e) ¹¹	Carbon removals					
	Corporate carbon offsets ⁷	-471,400	-324,100 ⁸	-167,000 ⁹	-70,000 ¹⁰	0
	Gross emissions (Scope 3)	15,570,000	20,280,000	23,020,000	22,260,000	24,460,000
	Manufacturing (purchased goods and services)	9,400,000	13,400,000	16,200,000	16,100,000	18,900,000
	Product transportation (upstream and downstream)	1,500,000	1,900,000	1,750,000	1,800,000	1,400,000
	Product use (use of sold products)	4,600,000	4,900,000	4,990,000	4,300,000	4,100,000
	End-of-life processing	70,000	80,000	80,000	60,000	60,000
	Carbon removals					
	Product carbon offsets	-13,500	0	-500,000 ¹²	0	0
	Total gross scope 3 emissions (corporate and product) (metric tons CO ₂ e)	15,980,000	20,545,800	23,128,400	22,550,000	24,980,000
Total gross carbon footprint (without offsets) (metric tons CO ₂ e) ¹³		16,100,000	20,600,000	23,200,000	22,600,000	25,100,000
Total net carbon footprint (after applying offsets) (metric tons CO ₂ e) ¹²		15,600,000	20,300,000	22,530,000	22,530,000	25,100,000

Notes:

- For data on years prior to 2019, please reference past Environmental Progress Reports.
- Totals might not add up due to rounding.

- Apple's carbon footprint boundary is aligned with the Greenhouse Gas (GHG) Protocol framework and includes emissions that are material and relevant to Apple, where data is available. Apple's carbon footprint includes direct scope 1 emissions; indirect scope 2 emissions from purchased electricity, steam, heating, and cooling; and indirect scope 3 emissions from purchased goods and services, transportation and distribution, business travel, employee commute, product use, and end of life.
- Apple is carbon neutral for corporate emissions as of April 2020. Beginning in fiscal year 2022, we've expanded our footprint boundary to include scope 3 emissions associated with work from home, third-party cloud services, electricity transmission and distribution losses, and upstream impacts from scope 1 fuels.
- Emissions from R&D processes and refrigerant leaks.
- We estimate the life cycle emissions associated with our use of renewable electricity for our corporate facilities to be about 70,000 metric tons CO₂e. We do not currently account for these emissions in our carbon footprint, due to the poor quality of this data.
- Beginning in fiscal year 2021, we're accounting for scope 2 emissions from the purchase of district heating, chilled water, and steam.
- Beginning in fiscal year 2020, we updated our methodology for calculating emissions from employee commute to reflect employees working from home during COVID-19.
- For a detailed breakdown of carbon offset purchases applied to our corporate footprint, see the carbon offsets table on the following page.

- We retired 324,100 metric tons of carbon credits from the Alto Mayo project in Peru and Chyulu Hills project in Kenya to maintain carbon neutrality for our corporate emissions in fiscal year 2022. This project is certified to the VCS and CCB standards.
- We retired 167,000 metric tons of carbon credits from the Chyulu Hills project in Kenya to maintain carbon neutrality for our corporate emissions in fiscal year 2021. This project is certified to the VCS and CCB standards.
- We retired 70,000 metrics tons of carbon credits — 53,000 from the Chyulu Hills project in Kenya and 17,000 from the Cispatá Mangrove project in Colombia.
- Because we're committed to accuracy and transparency, we regularly refine our product life cycle assessment model and sources of data. For example, last year we obtained more granular data summarizing in which countries our products are sold and used, resulting in more granularity possible for grid emission factors used in the carbon footprint of the product use phase. The net result was an increase in our fiscal year 2021 carbon footprint. When using the same level of data granularity and model as fiscal year 2021, our product use greenhouse gas emissions in fiscal year 2021 would have been about 2.5 percent lower.
- For fiscal year 2021, we retired credits from the Chyulu Hills project in Kenya and purchased carbon credits from two additional projects to offset a total of 500,000 metric tons of direct emissions across our value chain. The first project, a REDD+ coastal conservation project in Guatemala, protects and conserves forests from deforestation and degradation. The second project aims to establish forests on about 46,000 hectares of barren land that isn't otherwise in use across seven counties in the Guizhou province of China. Both projects are certified to the same high standards that we require for projects in the Restore Fund, including VCS and CCB standards.

- Due to rounding, our gross and net carbon footprints do not always equal the sum of the subtotals disclosed above.

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High quality carbon credits

We retired the following high-quality carbon credits toward our corporate emissions footprint for 2023.

Project name	Project description	Vintage	Volume retired (metric tons CO ₂ e)	Registry link
Chyulu Hills	The Chyulu Hills REDD+ Project (CHRP) is a multi-partner initiative designed to promote climate change mitigation and adaptation, restore biodiversity, and create alternative livelihoods under the UN scheme of Reducing Emissions from Deforestation and forest Degradation (REDD+). It's located in the Tsavo-Amboseli ecosystem in southeastern Kenya and stretches over an area of over 410,000 hectares. Its main geographic feature is the volcanic Chyulu Hills mountain range, from which the project derives its name. This project presents a broad ecosystem approach, including REDD+, to provide long-term sustainable financing and management to maintain the ecological integrity of an iconic African landscape. The project will help protect a very high-value wildlife and biodiversity area while supporting the development needs of Indigenous and other local communities.	2018	230,000	registry.terra.org/app/projectDetail/VCS/1408
Guinan	The Guinan Afforestation Project is located in the Guizhou Province of China and contributes to carbon removal and local sustainable development by planting trees on the barren lands. The project is planting across 46,000 ha on barren hills and degraded lands. The project activity aims to enhance biodiversity conservation by increasing the connectivity of forests, improve soil and water conservation, and generate income and job opportunities for local communities.	2019, 2020, and 2021	255,000	https://registry.terra.org/app/projectDetail/VCS/2070

Data

Carbon footprint by product

The following tables list the carbon footprints (in kilograms) of Apple products sold as of March 9, 2024, along with select configurations.¹

		Storage configurations				
iPhone	Unit	64GB	128GB	256GB	512GB	1TB
iPhone 15	kg CO ₂ e	–	56	61	74	–
iPhone 15 Plus	kg CO ₂ e	–	61	66	79	–
iPhone 15 Pro	kg CO ₂ e	–	66	71	83	107
iPhone 15 Pro Max	kg CO ₂ e	–	–	75	87	110
iPhone 14	kg CO ₂ e	–	61	67	83	–
iPhone 14 Plus	kg CO ₂ e	–	68	75	91	–
iPhone 13	kg CO ₂ e	–	64	71	83	–
iPhone SE (3rd generation)	kg CO ₂ e	46	50	58	–	–

		Storage configurations					
iPad	Unit	64GB	128GB	256GB	512GB	1TB	2 TB
iPad Pro 12.9-inch (6th generation) Wi-Fi + Cellular	kg CO ₂ e	–	135	142	156	183	284
iPad Pro 11-inch (4th generation) Wi-Fi + Cellular	kg CO ₂ e	–	100	107	121	148	249
iPad Air (5th generation) Wi-Fi + Cellular	kg CO ₂ e	80	84	92	–	–	–
iPad (10th generation) Wi-Fi + Cellular	kg CO ₂ e	72	82	–	–	–	–
iPad (9th generation) Wi-Fi + Cellular	kg CO ₂ e	75	–	84	–	–	–
iPad mini (6th generation) Wi-Fi + Cellular	kg CO ₂ e	68	71	77	90	–	–

		Select product configurations			
Apple Watch ²	Unit	Aluminum case with Sport Loop	Stainless steel case with Sport Loop	Titanium case with Alpine Loop	Titanium case with Trail Loop
Apple Watch Ultra 2	kg CO ₂ e	–	–	12	11
Apple Watch Series 9	kg CO ₂ e	8	29	–	–
Apple Watch SE	kg CO ₂ e	7	–	–	–

Note: Dashes indicate that the configuration does not exist.

- 1

Product carbon footprint data for Apple products are published in our Product Environmental Reports and are accurate as of product launch. In instances where carbon models were developed prior to product launch, we use preproduction units.
- 2

Greenhouse gas emissions prior to applied high quality carbon credits are represented for all Apple Watch models with the exception of stainless steel Apple Watch Series 9 paired with sport loop.